AI-POWERED HR TECHNOLOGY

AI-Based Employee Skill Gap & Training Recommendation System

SkillSync AI

A workforce intelligence platform to identify capability gaps and guide employees toward relevant learning.

PROJECT TEAM

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The problem we are solving

Why workforce skill visibility matters

Today’s challenge

- Employee skills are scattered across profiles, assessments and spreadsheets.
- Managers may not know which competencies are below role expectations.
- Generic training can waste time when it does not target actual skill gaps.
- Without clear tracking, workforce readiness is difficult to measure.

Our approach

- Create a shared view of employee skills and role requirements.
- Compare current proficiency with expected proficiency.
- Prioritize the gaps that matter most for each role.
- Recommend learning content that matches identified gaps.

Proposed solution: SkillSync AI

One workflow from employee skills to targeted development

01

Collect

Employee profile & skill levels

02

Compare

Current skills vs. role requirements

03

Identify

Calculate gaps & readiness

04

Recommend

Rank relevant training courses

05

Improve

Track learning & progress

OUTCOME **A clearer picture of workforce readiness and a practical path to close skill gaps.**

Core features

The prototype brings key HR workflows into one dashboard

Workforce overview

KPIs for employees, assessed skills, identified gaps and training completion.

Employee directory

Search employees, filter by department and open individual profiles.

Skill gap analysis

Compare proficiency with role requirements and calculate readiness.

Training recommendations

Surface courses relevant to an employee’s missing skills.

Course enrollment

Explore learning options and record employee course enrollment.

Reports & insights

Visualize competency levels and department readiness.

System architecture

Frontend, application API, AI recommendation service and database

USER / HR ADMIN

Uses dashboard to manage employees, view gaps and explore training.

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REACT + VITE FRONTEND

Responsive dashboard • navigation • employee views • charts • course interactions

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JAVA SPRING BOOT API

REST endpoints for employees and skill analysis; primary application backend

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PYTHON FASTAPI SERVICE

TF-IDF vectorization + cosine similarity to score course relevance

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MYSQL DATABASE

Profiles • skills • role requirements • courses • enrollments • recommendations

How the recommendation engine works

Content-based matching using TF-IDF and cosine similarity

Input

- Employee skill gaps (skill name + gap level).
- Available courses (title, description and associated skills).
- Only positive skill gaps are considered.

Matching pipeline

1	**Build gap text**	Repeat skill terms based on gap level
2	**Vectorize**	Convert text into TF-IDF vectors
3	**Compare**	Calculate cosine similarity
4	**Rank**	Sort courses by relevance score

Higher similarity = stronger text match to the employee's identified gaps.

Example: skill gap analysis

Illustrative demo data from the project's analysis endpoint

Skill	Current	Required	Gap
Java	4	4	**0**
Python	2	4	**2**
SQL	3	4	**1**
React	1	3	**2**

75%

Readiness score

Calculated from average capped current-to-required proficiency ratios.

Example gaps: Python (2 levels), SQL (1 level), React (2 levels).

Technology stack

Technologies present in the supplied project

Frontend	React 18 • Vite • JavaScript • CSS • Recharts • Lucide icons
Application backend	Java • Spring Boot • REST controllers • Maven
AI recommendation API	Python • FastAPI • scikit-learn • TF-IDF • cosine similarity
Data layer	MySQL • relational schema • foreign keys • constraints
Development	Node.js / npm • Python environment • Java 17+ • Maven

Database design

MySQL schema supports the employee-to-training lifecycle

users	Identity, email, password hash and access role
employees	Employee code, job role, manager and experience
skills	Skill name and category
employee_skills	Employee proficiency and verification
job_roles + role_required_skills	Role expectations and required skill levels
courses + course_skills	Course metadata and skills taught
enrollments	Enrollment status and learning progress
skill_assessments + recommendations	Assessment history and recommendation scores

Typical user journey

How an HR administrator can use the prototype

1. **Open overview** — Review workforce KPIs and department readiness.
2. **Select employee** — Search or filter the employee directory.
3. **Analyze skills** — Compare proficiency against role requirements.
4. **Review courses** — Use relevance-ranked training recommendations.
5. **Enroll & follow up** — Record course enrollment and monitor progress.

Current implementation & next steps

The supplied ZIP describes a hackathon-ready starter / MVP scaffold

Included in the project

- Interactive React dashboard with demo data and navigation.
- Employee listing, filters and skill analysis UI.
- Spring Boot REST API starter endpoints.
- FastAPI recommendation endpoint using TF-IDF similarity.
- MySQL schema with sample departments, skills, roles and courses.

Recommended future work

- Connect frontend screens to persistent backend APIs.
- Implement secure authentication, authorization and validation.
- Replace demo calculations with database-driven role and skill data.
- Add employee self-assessments and manager verification.
- Track course completion and measurable skill improvement.

CONCLUSION

Build skills.
Close gaps.
Grow together.

SkillSync AI turns workforce skill data into focused learning recommendations—helping teams make development more measurable and role-aligned.

THANK YOU

Questions?

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